Pick columns, do maths

In Ep 04 you picked only the columns you need, then let SQL do the maths on every row. Here is every line, with what each part means, so you can type along. You need the **orders** table from Ep 03 (ep04_columns_and_maths.sql makes it if you missed it).

Column names
Write them instead of the star, and only those columns come back.

AS
Gives a column a proper name. Like a nickname.

*** (the star)**
Two jobs. Right after SELECT: every column. Between two numbers: multiply.

ROUND
`ROUND(x, 2)` keeps 2 numbers after the decimal point.

Before step 1 `make sure you're in shop`

```
1  USE shop;
```

Open Workbench, click your connection, type it and run it.

`✓ use shop · 0 row(s) affected`

Step 1 · pick your columns `names instead of the star`

```
1  SELECT city, amount FROM orders;
```

Instead of the star, write the names of the columns you want. A real company table can have 10, 20, even 50 columns, but most of the time you only need two. Run it, and only those two come back:

city	amount
london	12.99
leeds	4.50
hatfield	21.50

Same comma rule as last time: a comma between the names, none after the last one. And you choose the order: `SELECT amount, city` puts amount first.

It's like your marksheet. If someone only asks for your name and your maths marks, you don't hand over the whole sheet. You pick those two columns. Added order 1004 in Ep 03's your turn? Then you'll see a 4th row everywhere on this sheet. That's fine.

Step 2 · maths SQL works it out on every row

```sql
SELECT order_id,
  amount + 2
FROM orders;
```

Delivery is 2 pounds on every order. SQL adds 2 to every row at once: 12.99 becomes 14.99. Your table doesn't change, SQL only shows you the answer.

`14.99 · 6.50 · 23.50`

```sql
SELECT order_id,
  amount + 2 AS total
FROM orders;
```

The column was called amount + 2. Looks messy. AS gives it a proper name, so now it says total.

`order_id · total`

```sql
SELECT order_id,
  amount * 0.9 AS sale_price
FROM orders;
```

Diwali sale: 10% off. Costs 100, you pay 90, so it's times 0.9. Between two numbers, the star means multiply.

`11.691 · 4.050 · 19.350`

```sql
SELECT order_id,
  ROUND(amount * 0.9, 2)
  AS sale_price FROM orders;
```

11.691 has three numbers after the decimal point, but money has two. ROUND, bracket, our maths, comma 2, close the bracket.

`11.69 · 4.05 · 19.35`

+	add
-	take away
*	multiply
/	divide

Break it on purpose

```sql
SELECT city amount FROM orders;
```

No error. But look: the column says amount, and it's full of cities. Without the comma, SQL thinks amount is a nickname for city, just like `city AS amount`. If your columns look weird, check your commas.

amount
london
leeds
hatfield

Got an error? unknown column

Error 1054 · Unknown column 'amout'

You spelled it wrong: `amout` instead of `amount`. Fix the spelling and run it again. The name has to match the table exactly.

Your turn

Say **£1 = ₹125**. Show every order in rupees, and call the column **rupees**. Then comment the rupee price of **order 1003** on the reel.

Next, on Wednesday: **WHERE, AND, OR, IN**. How to pick only the rows you want.

Answer: SELECT order_id, amount * 125 AS rupees FROM orders; gives 1623.75, 562.50 and 2687.50, so order 1003 is ₹2687.50.